Web Data Integration
Final Report

Pharmaceutical Drug Integration: Linking US, EU and Canadian Approval Data

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1 Use Case

This project builds a unified view of approved pharmaceutical drugs across the United States, European Union, and Canada by integrating three official regulatory datasets: Drugs@FDA (US), EMA Medicines (EU), and the Health Canada Drug Product Database (DPD). The central entity is a medicine identified by its brand name and active substance(s); each integrated record includes licence holder, regulatory status, product class, ATC code, dosage form, route, strengths, substance count, regional dates, and generic/biosimilar/orphan flags.

2 Requirements Suitability

2.1 Schema and basic profile

Table 1: Overview of the input datasets

Dataset	Source	Format	Rows	Entities	Attr.
Drugs@FDA	U.S. FDA	TXT/TSV	51,839	8,175	56
EMA Med.	EMA	XLSX/CSV	2,746	2,648	39
DPD	Health Can.	TXT/CSV	42,042	41,636	87

Table 1 summarises the three input datasets. Drugs@FDA contains 51,839 product rows forming 8,175 entities across 29,377 applications (78.3% ANDAs). EMA Medicines contains 2,746 rows consolidated into 2,648 entities (including 28 records recovered via INN fallback), predominantly human medicines (85.6%). The Health Canada DPD contains 58,137 product records consolidated into 41,636 entities. High missingness in all three datasets is largely structural: optional fields, category-specific attributes, or regulatory-event dates that apply only to specific statuses.

2.2 Integrated schema and overlap with input schemata

The integrated schema contains 16 attributes (Table 2); eleven occur in at least two sources. `activeSubstances` and `strengths` are list-valued, while the remaining 14 attributes are single-valued (including the numeric attribute `activeSubstanceCount`). Strengths are represented as numeric value–unit pairs.

2.3 Entity Identification and Count Requirement

A medicine is identified by its `brandName` and `activeSubstances`. Contrary to initial assumptions, `licenceHolder` cannot be used as an identifying attribute because the same medicine is frequently marketed by different regional subsidiaries or unrelated licensing partners across jurisdictions (for example, *Verkazia* is held by `HARROW EYE` in the US, `Santen Oy` in the EU, and `APOTEX INC` in Canada).

To estimate how many medicines appear in multiple datasets based on exact matching, we build a normalized key (`medicine_id`) from `brandName` and `activeSubstances` by

Table 2: Attribute Intersection with Integrated Schema

Attribute name	Attribute type	Datasets in which the attribute is found
`brandName`	string	Drugs@FDA, EMA Medicines, DPD
`activeSubstances`	list (string)	Drugs@FDA, EMA Medicines, DPD
`activeSubstanceCount`	numeric (integer)	Drugs@FDA (derived), EMA Medicines (derived), DPD
`licenceHolder`	string	Drugs@FDA, EMA Medicines, DPD
`regulatoryStatus`	string (categorical)	Drugs@FDA, EMA Medicines, DPD
`productClass`	string (categorical)	Drugs@FDA (derived), EMA Medicines, DPD
`atcCode`	string	EMA Medicines, DPD
`generic`	boolean	Drugs@FDA (derived), EMA Medicines
`biosimilar`	boolean	EMA Medicines
`orphanMedicine`	boolean	EMA Medicines
`dosageForm`	string	Drugs@FDA, DPD
`route`	string	Drugs@FDA, DPD
`strengths`	list (numeric value, unit)	Drugs@FDA, DPD
`usApprovalDate`	date	Drugs@FDA
`euAuthorisationDate`	date	EMA Medicines
`caStatusDate`	date	DPD

converting text to upper case, removing accents and dosage strengths, standardizing punctuation, and sorting multiple active substances. Let F, E and D denote the sets of `medicine_id` values in Drugs@FDA, EMA Medicines and the DPD. The pairwise exact overlaps are $|F \cap E| = 428$, $|F \cap D| = 949$ and $|E \cap D| = 594$, yielding a deduplicated union of $|(F \cap E) \cup (F \cap D) \cup (E \cap D)| = 1{,}407$ distinct medicines – well above the required minimum of 1,000. (Note that this initial exploratory estimate was calculated on raw tabular extracts prior to final EMA entity consolidation; when evaluated via transitive graph clustering over the finalized XML datasets in Section 4.5, the exact baseline resolves to 1,372 unified multi-source medicine clusters).

However, exact matching misses many identical medicines because both identifying attributes can vary across countries (Table 3):

- **Active substance naming and salt forms:** The same substance can have different regional spellings (e.g., `CYCLOSPORINE` in the US and Canada vs. `ciclosporin` in the EU for *Verkazia*) or different chemical and salt representations (e.g., `GLYCOPYRROLATE` in the US, `glycopyrronium bromide` in the EU, and `GLYCOPYRRONIUM (GLYCOPYRRONIUM BROMIDE)` in Canada for *Seebri*).
- **Brand name suffixes:** The same medicine may carry different device or formulation suffixes across markets, such as `SEEBRI NEOHALER` in Drugs@FDA versus `Seebri Breezhaler` in EMA Medicines and the DPD.

3 Phase I: Data Selection and Translation

Phase I translates three pharmaceutical datasets into a shared XML target schema using Altova MapForce 2027. The outputs are validated XML files consumed by Phases II and III.

Table 3: Examples of identical medicines with differing identifying attributes across sources

Dataset	Brand Name	Active Substance
Drugs@FDA	VERKAZIA	CYCLOSPORINE
EMA Medicines	Verkazia	ciclosporin
Health Canada DPD	VERKAZIA	CYCLOSPORINE
Drugs@FDA	SEEBRI NEOHALER	GLYCOPYRROLATE
EMA Medicines	Seebri Breezhaler	glycopyrronium bromide
Health Canada DPD	SEEBRI BREEZHALER	GLYCOPYRRONIUM (GLYCOPYRRONIUM BROMIDE)

3.1 Target Schema

The target schema (`medicines.xsd`) represents one medicine entity per source record, identified by `brandName` and `activeSubstances`, with 16 attributes and two metadata fields (`id`, `source`). Three conventions apply across all mappers: missing values are omitted rather than written as empty tags; repeating values use typed wrapper elements (`activeSubstances`, `strengths`); and all dates follow ISO 8601 (YYYY-MM-DD). Controlled vocabularies are enforced for `regulatoryStatus` (six values: *Authorised, Suspended, Pending, Withdrawn, Application withdrawn, Refused*) and `productClass` (*Human, Veterinary, Disinfectant*).

3.2 Shared Helper Functions

A shared MapForce library (`wdi_helpers.mfd`) provides ten user-defined functions imported by all mappers (Table 4). Lookup XML files cover 25 regulatory statuses, 227 dosage forms, 147 routes, and 202 strength units across all three sources.

Table 4: Shared UDFs used across mappers

Function	Purpose
`makeMedicineId`	Constructs a stable entity ID from source prefix and record key
`getActiveSubstances`	Tokenizes substance string; supports INN fallback for blank primary field
`standardizeRegulatoryStatus`	Maps raw status values to the six controlled vocabulary entries
`standardizeProductClass`	Maps category values to Human / Veterinary / Disinfectant
`standardizeAtcCodes`	Combines human and vet ATC codes; deduplicates, sorts, corrects typos
`yesNoToBoolean`	Converts Yes/No strings to `xs:boolean`; omits element when false
`parseAuthorisationDate`	Converts DD/MM/YYYY to ISO 8601 YYYY-MM-DD
`stdDosageForm` / `dosageFormList`	Standardizes single or multi-value dosage form strings
`stdRoute` / `routeList`	Standardizes single or multi-value administration routes
`splitStrengths`	Tokenizes a DPD strength string into (value, unit) pairs

3.3 EMA Medicines Mapper

The EMA dataset originates from a flat table of 2,746 rows consolidated into 2,648 unique medicine entities (`ema_entities.csv`). Each consolidated entity row maps to one output `<medicine>` element. Because EMA does not publish strength or dosage form data in the register export, `dosageForm`, `route`, and `strengths` are structurally absent from EMA output. Table 5 lists all field correspondences.

Table 5: EMA Medicines field mapping

Target field	Source column	Transformation
`id`	medicine_id	`makeMedicineId("EMA", id)`
`brandName`	Name of medicine	Direct copy
`activeSubstances`	Active substance / INN	`getActiveSubstances`; INN fallback
`activeSubstanceCount`	(derived)	`count(activeSubstances)`
`licenceHolder`	MA holder	Direct copy
`regulatoryStatus`	Medicine status	`standardizeRegulatoryStatus`
`productClass`	Category	`standardizeProductClass`
`atcCode`	ATC code (human) + ATCvet	`standardizeAtcCodes`
`generic` / `biosimilar` / `orphanMedicine`	Generic / Biosimilar / Orphan	`yesNoToBoolean`
`euAuthorisationDate`	Marketing authorisation date	`parseAuthorisationDate`

The most intricate correspondence handles active substances. For 57 of the 2,746 rows, both `Active substance` and the `INN / common name` column are blank; because `activeSubstances` is mandatory in the target schema, these rows are excluded via a row filter. A further 28 rows have a blank primary field but a populated INN column: the mapper applies an INN fallback, normalizing whitespace and selecting the non-empty value. Commas and the word "and" are not treated as separators, preserving multi-word biological descriptions. Consolidating the 2,689 usable rows by normalized brand name and active substance set yields **2,648 unique medicine entities** (`ema_entities.csv`).

EMA maintains separate human and veterinary ATC code columns. `standardizeAtcCodes` concatenates both, strips asterisks used for provisional codes, drops the *Not yet assigned* placeholder, corrects a known EMA typo (`VO4D` → `V04D`), then deduplicates, sorts, and joins codes with `"; "`. Date conversion (`parseAuthorisationDate`) reorders DD/MM/YYYY to YYYY-MM-DD. The three boolean flags are stored as Yes/No strings; `yesNoToBoolean` converts them and omits the element when false, consistent with the `minOccurs="0"` schema convention. The full export passed XSD validation with **2,648 unique `EMA_` IDs** and zero duplicate or schema errors.

3.4 Drugs@FDA Mapper

The Drugs@FDA source consists of relational TSV tables in which multiple product rows across applications (`ApplNo`) correspond to the same medicine entity. Product records from `Products.txt` are joined to `Applications.txt` on `ApplNo`, linked to marketing status lookup values, and consolidated by normalized `brandName` (`DrugName`) and sorted `activeSubstances` (`ActiveIngredient`) into **8,175 unique medicine entities**. Eleven orphan product rows lacking a matching application and records with empty active ingredients (such as `ApplNo=221337`, `MEDICAL AIR, USP`) are excluded during

preparation. Table 6 summarises the field mappings.

Table 6: Drugs@FDA field mapping

Target field	Source column	Transformation
`id`	medicine_id	`makeMedicineId("FDA", id)`
`brandName`	DrugName (plurality)	Direct copy
`activeSubstances`	ActiveIngredient	`getActiveSubstances`; separator `"; "`
`activeSubstanceCount`	(derived)	`count(activeSubstances)`
`licenceHolder`	SponsorName (plurality)	Direct copy
`regulatoryStatus`	MarketingStatusID	`standardizeRegulatoryStatus`
`productClass`	(constant)	`"Human"`
`generic`	ApplType (`ANDA`/`NDA`/`BLA`)	`true` if all `ANDA`; `false` if all `NDA`/`BLA`; omit if mixed
`dosageForm`	Form (form component)	`fdaFormPart` → `dosageFormList`
`route`	Form (route component)	`fdaRoutePart` → `routeList`
`strengths`	Strength	Pre-split combination tokens → `splitStrengths`
`usApprovalDate`	SubmissionStatusDate (`AP`)	Earliest non-empty approval date (ISO 8601)

Drugs@FDA stores dosage form and administration route jointly in `Products.Form` (e.g. `"TABLET;ORAL"`); the helper functions `fdaFormPart` and `fdaRoutePart` split this composite field before lookup standardization. Combination strength strings such as `600MG/90MG` are pre-split into individual mass tokens prior to numeric value/unit extraction. Because Drugs@FDA covers only human drugs, `productClass` maps to *Human*. The `generic` flag is derived from application types: `true` when all contributing applications are `ANDA`, `false` when all are innovator applications (`NDA` or `BLA`), and omitted for the approximately 375 mixed entities. The resulting XML export (`fda_medicines.xml`) passed XSD and duplicate validation with **8,175 unique** `FDA_` **IDs**.

3.5 Health Canada DPD Mapper

The DPD source is a pre-consolidated CSV of **41,636 medicine entities** derived from 58,137 product records across 12 Health Canada tables (`dpd_entities.csv`). Unlike the EMA flat table, each DPD entity already aggregates multiple product records, so the mapper operates on one row per entity without further consolidation. The DPD source provides all 16 target attributes including `dosageForm`, `route`, and `strengths`, making it the most complete of the three sources.

The main mapping challenge in DPD is strength parsing. Each entity row carries a semicolon-delimited strength string (e.g. `"10MG; 0.5%"`); `splitStrengths` tokenizes this into individual tokens, from which `strengthValue` extracts the numeric component and `strengthUnit` normalizes the unit via `lookups/strength_unit.xml`. Tokens without a parseable numeric value are filtered out before the `<strength>` wrapper is created, and the wrapper itself is omitted when no valid tokens remain. Regulatory status presents a similar aggregation challenge: DPD entities can carry multiple historical statuses (e.g. `"MARKETED; CANCELLED POST MARKET"`), so `standardizeRegulatoryStatus` selects the highest active status according to the precedence hierarchy Authorised > Suspended > $\cdots$ > Refused.

Table 7: Health Canada DPD field mapping

Target field	Source column	Transformation
`id`	medicine_id	`makeMedicineId("DPD", id)`
`brandName`	brand_name	Direct copy
`activeSubstances`	active_substances	`getActiveSubstances`; separator `"; "`
`activeSubstanceCount`	(derived)	`count(activeSubstances)`
`licenceHolder`	licence_holder	Direct copy
`regulatoryStatus`	current_status	`standardizeRegulatoryStatus`
`productClass`	product_class	`standardizeProductClass`
`atcCode`	atc_code	`standardizeAtcCodes`
`biosimilar`	biosimilar	`yesNoToBoolean`
`dosageForm`	dosage_forms	`dosageFormList`
`route`	routes	`routeList`
`strengths`	strengths	`splitStrengths` → (value, unit) pairs
`caStatusDate`	status_date	Direct copy (ISO date)

The full export passed XSD validation with **41,636 records**, 100,094 strength pairs across 41,574 medicines, and no schema errors.

3.6 Integrated XML Validation Summary

All three translated XML files in `data/4_translated_mapforce/` were validated against `medicines.xsd` and checked for duplicate IDs, duplicate attribute content, and duplicate normalized entity keys using `check_schema.py` (Table 8). Together, the three sources provide **52,459 validated medicine entities** for Phase II identity resolution.

Table 8: Validation summary of translated XML datasets

Validation check	`dpd_medicines.xml`	`ema_medicines.xml`	`fda_medicines.xml`
Records	41,636	2,648	8,175
Unique IDs	41,636	2,648	8,175
XSD Schema valid	True	True	True
Duplicate IDs	0	0	0
Duplicate content (excl. ID)	0	0	0
Duplicate normalized entities	0	0	0

4 Phase II: Identity Resolution

Phase II resolves cross-jurisdictional medicine correspondences across Health Canada DPD (41,636 entities), EMA Medicines (2,648 entities), and Drugs@FDA (8,175 entities). Evaluating the full Cartesian product across the three dataset pairs would require $41{,}636 \times 2{,}648 + 41{,}636 \times 8{,}175 + 2{,}648 \times 8{,}175 = 472{,}273{,}828$ pairwise comparisons, motivating blocking, supervised and zero-shot matchers, and transitive connected-components clustering.

4.1 Gold-Standard Dataset Construction and Stratified Split

To construct a representative benchmark that tests both standard matches and adversarial pharmaceutical corner cases, candidate pairs for manual annotation were sampled

in two balanced pools using `gold_standard_pair_gen.py`:

- **Ordinary Pool (`hard=0`):** Contains candidate pairs with exact or high-similarity normalized brand and active substance keys alongside random negative pairs.
- **Hard Pool (`hard=1`):** Targets domain-specific corner cases where string similarity is misleading: (i) identical brand names with different active substances or combination vs. single-agent formulas, (ii) identical active substances marketed by competing generic manufacturers, and (iii) same medicine entities with regional INN salt variations or device/formulation suffixes.

Three team annotators manually reviewed all candidate batches across `DPD/EMA` (1,000 pairs), `DPD/FDA` (769 pairs), and `EMA/FDA` (1,729 pairs), yielding **3,498 unique reviewed pairs** with zero duplicate rows, zero conflicting labels, and zero unlabeled records. Two records were labeled as a match (`match` = 1) if and only if they share the same pharmacological active substance set (accounting for salt forms, hydration states, and regional INN spellings) and represent the same commercial medicine entity (including regional subsidiaries of the same corporate group, while excluding unrelated competing generic manufacturers).

Using `gold_standard_combine_split.py`, the 3,498 unified pairs were partitioned into an 85% development split (`gs_pair_train_val.csv`, **2,973 pairs**) and a 15% held-out test split (`gs_pair_test.csv`, **525 pairs**) stratified jointly by (`source_pair`, `hard`, `match`) with fixed seed 42 (Table 9).

Table 9: Stratified gold-standard dataset distribution across source pairs, difficulty pools, and partitions

Source Pair	Difficulty Pool	Label	Whole	Train/Val (85%)	Test (15%)
DPD / EMA	Ordinary (`hard=0`)	Match (1)	198	168	30
DPD / EMA	Ordinary (`hard=0`)	Non-match (0)	302	257	45
DPD / EMA	Hard (`hard=1`)	Match (1)	217	184	33
DPD / EMA	Hard (`hard=1`)	Non-match (0)	283	241	42
DPD / FDA	Ordinary (`hard=0`)	Match (1)	122	104	18
DPD / FDA	Ordinary (`hard=0`)	Non-match (0)	364	309	55
DPD / FDA	Hard (`hard=1`)	Match (1)	126	107	19
DPD / FDA	Hard (`hard=1`)	Non-match (0)	157	133	24
EMA / FDA	Ordinary (`hard=0`)	Match (1)	282	240	42
EMA / FDA	Ordinary (`hard=0`)	Non-match (0)	670	570	100
EMA / FDA	Hard (`hard=1`)	Match (1)	292	248	44
EMA / FDA	Hard (`hard=1`)	Non-match (0)	485	412	73
Total	**All Pools**	**All Labels**	**3,498**	**2,973**	**525**

4.2 Blocking and Candidate Generation

The unreduced search space across the three translated datasets encompasses 41,636 × 2,648 = 110,252,128 pairs for DPD–EMA, 41,636 × 8,175 = 340,374,300 pairs for DPD–FDA, and 2,648 × 8,175 = 21,647,400 pairs for EMA–FDA, yielding a total Cartesian product of **472,273,828 possible comparisons**. Performing detailed attribute matching or transformer inference across nearly half a billion pairs is computationally prohibitive. The goal of blocking is to filter out obvious non-matches while maintaining

high completeness on true correspondences. Performance is governed by two complementary metrics:

- **Reduction Ratio (RR):** $\text{RR} = 1 - \frac{|\mathcal{C}|}{|A \times B|}$, measuring the relative reduction in pairwise comparisons (target: $> 99.5\%$).
- **Pairs Completeness (PC):** $\text{PC} = \frac{|\mathcal{C} \cap \mathcal{M}|}{|\mathcal{M}|}$, the fraction of true matches $\mathcal{M}$ retained in candidate shortlist $\mathcal{C}$ (target: $> 97.0\%$ per source pair).

Lexical Blocking Baselines. We initially benchmarked standard lexical blocking rules implemented via PyDI (`StandardBlocker` and `TokenBlocker`) on the development split (`gs_pair_train_val.csv`, 1,038 unique eligible positive matches after EMA ID alignment and deduplication; Table 10):

1. **Brand Prefix-4:** Compares the first four alphanumeric characters of normalized brand names. While achieving high reduction ratios ($> 99.96\%$), it missed 39 of the 352 DPD–EMA matches (PC = 88.92%) due to regional brand prefixes (e.g., `TEVA-`, `APO-`, or formulation prefixes).
2. **Brand Trigrams (capped at 1,000):** Tokenizes brand names into overlapping 3-character shingles. To prevent Cartesian explosions on frequent pharmaceutical fragments (e.g., `"TAB"`, `"SOL"`), candidate generation was capped at $a \times b \leq 1{,}000$ co-occurrences per shingle. However, trigrams alone missed 39 matches on DPD–FDA (PC = 81.52%).
3. **Prefix-4 $\cup$ Trigrams:** Combining both lexical rules improved completeness to $\geq 98.30\%$ on all pairs, but generated 1,369,537 candidates (663,705 on DPD–FDA alone), representing a substantial downstream processing burden.

Dense Semantic Embeddings (Qwen3-Embedding-0.6B). To capture semantic equivalence across international nonproprietary names (INNs), salt variations, and brand synonyms, we evaluated dense vector blocking using `Qwen/Qwen3-Embedding-0.6B` (1,024-dimensional representations). For each medicine entity, the normalized brand name and alphabetically sorted active substances were concatenated and encoded without fine-tuning. Cosine nearest-neighbor search was executed from the smaller source into the larger source. A shortlist of the top-5 nearest neighbors per medicine produced only **67,355 total candidates** while retaining 1,021 of 1,038 development positives (**98.36% pooled completeness**; Table 10).

Frozen Evaluation and Final Matcher Handoff. When Qwen Top-5 was evaluated on the held-out test set (185 known matches), it achieved 98.41% on DPD–EMA and 97.30% on DPD–FDA, but dropped to 92.94% (79/85) on EMA–FDA (95.68% pooled completeness), failing the per-source 97% target. Diagnostic error inspection revealed that missed EMA–FDA pairs shared brand prefixes that lexical prefix matching reliably captures.

Table 10: Comparison of lexical and dense embedding blocking methods on the development set (1,038 positive matches)

Source Pair	Blocking Method	Candidates	Matches	Retained	Completeness	Red. Ratio
DPD–EMA	Brand Prefix-4	16,398	352	313	88.92%	99.9851%
DPD–EMA	Brand Trigrams (cap 1,000)	431,353	352	323	91.76%	99.6088%
DPD–EMA	Prefix-4 ∪ Trigrams	443,207	352	346	98.30%	99.5980%
DPD–EMA	Qwen Top-5	13,240	352	346	98.30%	99.9880%
DPD–EMA	**Qwen Top-5 ∪ Prefix-4**	**27,966**	**352**	**350**	**99.43%**	**99.9746%**
DPD–FDA	Brand Prefix-4	133,069	211	210	99.53%	99.9609%
DPD–FDA	Brand Trigrams (cap 1,000)	544,036	211	172	81.52%	99.8402%
DPD–FDA	Prefix-4 ∪ Trigrams	663,705	211	210	99.53%	99.8050%
DPD–FDA	Qwen Top-5	40,875	211	210	99.53%	99.9880%
DPD–FDA	**Qwen Top-5 ∪ Prefix-4**	**162,930**	**211**	**211**	**100.00%**	**99.9521%**
EMA–FDA	Brand Prefix-4	4,456	475	468	98.53%	99.9794%
EMA–FDA	Brand Trigrams (cap 1,000)	261,601	475	472	99.37%	98.7915%
EMA–FDA	Prefix-4 ∪ Trigrams	262,625	475	474	99.79%	98.7868%
EMA–FDA	Qwen Top-5	13,240	475	465	97.89%	99.9388%
EMA–FDA	**Qwen Top-5 ∪ Prefix-4**	**15,934**	**475**	**474**	**99.79%**	**99.9264%**

To guarantee compliance with the project's $\geq 97\%$ completeness requirement across all three pairs, the final handoff rule was defined as the union:

$$\mathcal{C}_{\text{final}} = \mathcal{C}_{\text{Qwen top-5}} \cup \mathcal{C}_{\text{Prefix-4}}$$

As shown in Table 11, this hybrid rule recovers 6 additional test matches, bringing EMA–FDA to **97.65% completeness** and achieving **100.00% completeness** on DPD–EMA and DPD–FDA (**98.92% pooled completeness**, 183/185). The candidate shortlist totals **206,830 unique pairs**, compressing the search space by **99.9562%** (over 2,280× reduction). All candidates were persisted to `data/7_blocking_candidates/` with cryptographic SHA-256 checksums verified in `manifest.json`.

Table 11: Final candidate handoff summary and held-out test set evaluation (185 known matches)

Source Pair	Cartesian Space	Candidates	Red. Ratio	Test Matches	Retained	Completeness
DPD–EMA	110,252,128	27,966	99.9746%	63	63	**100.00%**
DPD–FDA	340,374,300	162,930	99.9521%	37	37	**100.00%**
EMA–FDA	21,647,400	15,934	99.9264%	85	83	**97.65%**
Pooled	**472,273,828**	**206,830**	**99.9562%**	**185**	**183**	**98.92%**

4.3 Rule-Based, Machine Learning, and Fine-Tuned PLM Matchers

4.4 Zero-Shot LLM Matcher Benchmark

To evaluate whether a pretrained Large Language Model can resolve cross-jurisdictional pharmaceutical entities without task-specific training examples, we implemented a strictly blind Zero-Shot LLM matcher (`src/identity_resolution/llm_zero_shot_matcher.py`) using Google's `gemini-3.8-flash` model via the official `google-genai` SDK (`temperature` = 0.0, structured JSON schema output enforcing integer `match` $\in \{0, 1\}$ and clinical rationale string `reason`).

Decontaminated Zero-Shot Prompt Architecture. Each test pair in `gs_pair_test.csv` ($n = 525$) is formatted with four record attributes per jurisdiction: `brandName`, `activeSubstances`, `licenceHolder`, and `dosageForm` (with missing values formatted as `"Not specified"`). The system instruction encodes three general domain rules without exposing any test-set drug names or few-shot examples:

1. **Active Substances:** Ingredients must represent the same pharmacological moiety; salt forms (e.g. hydrochloride, sodium, maleate, besylate), hydration states, and INN vs. USAN spellings (e.g. ciclosporin vs. cyclosporine) are treated as equivalent, whereas single-substance vs. multi-substance combination products are non-matches.

2. **Brand Name and Licence Holder:** Identical brands or minor formulation suffixes from regional subsidiaries of the same corporate group are matches, whereas generic products from unrelated competing manufacturers are non-matches.

3. **Dosage Form Variations:** Minor dosage form differences do not invalidate a match when brand name, active substance set, and corporate lineage agree.

Comparison Against the Naive Exact-Match Baseline. We benchmark the Zero-Shot LLM matcher against a deterministic **Naive Exact-Match Baseline** that predicts `match` = 1 if and only if normalized `brandName1` equals `brandName2` and their normalized sets of semicolon-delimited `activeSubstances` are identical. Table 12 reports overall test-set performance across all 525 held-out pairs.

Table 12: Held-out test set evaluation ($n = 525$ pairs): Naive Exact-Match Baseline vs. Zero-Shot LLM (`gemini-3.8-flash`)

Metric	Exact-Match Baseline	Zero-Shot LLM (`gemini-3.8-flash`)	Improvement (Δ)
Macro F1 Score	0.7307 (73.07%)	**0.9485 (94.85%)**	**+21.78 pp**
Match Class F1 (1)	0.6122 (61.22%)	**0.9344 (93.44%)**	**+32.21 pp**
Non-Match Class F1 (0)	0.8492 (84.92%)	**0.9626 (96.26%)**	**+11.34 pp**
Overall Accuracy	0.7829 (78.29%)	**0.9524 (95.24%)**	**+16.95 pp**
Macro Precision	0.8016 (80.16%)	0.9443 (94.43%)	+14.27 pp
Macro Recall	0.7154 (71.54%)	0.9534 (95.34%)	+23.80 pp
Match Precision (Class 1)	0.8333 (83.33%)	0.9128 (91.28%)	+7.95 pp
Match Recall (Class 1)	0.4839 (48.39%)	0.9570 (95.70%)	+47.31 pp
True Positives (TP)	90	**178**	+88 matches found
False Negatives (FN)	96	**8**	−88 missed matches
False Positives (FP)	18	**17**	−1 false alarm
True Negatives (TN)	321	**322**	+1

The Naive Exact-Match Baseline achieves a Macro F1 of 73.07% and misses more than half of all true matches ($\text{Recall}_1 = 48.39\%$, FN = 96) because regional salt modifiers and brand suffixes break exact string equality. By contrast, `gemini-3.8-flash` achieves **94.85% Macro F1** (95.24% accuracy), recovering 88 additional true matches and reducing false negatives by 91.7% (from 96 to 8).

Performance by Difficulty Pool and Source Jurisdiction Pair. Table 13 disaggregates test-set performance across difficulty pools and regulatory pair combinations. On the adversarial **Hard Pool** (`hard` = 1, 235 pairs), the Exact-Match Baseline fails completely on positive matches (Match F1 = 0.0000, Macro F1 = 0.3717), whereas the Zero-Shot LLM maintains **0.9296 Macro F1** (0.9167 Match F1, 93.19% accuracy). Across source pairs, the LLM achieves 0.9864 Macro F1 on `DPD/EMA`, 0.9704 on `DPD/FDA`, and 0.9144 on `EMA/FDA`.

Table 13: Zero-Shot LLM vs. Naive Baseline breakdown by difficulty pool and source jurisdiction pair

Subgroup	Pairs	Base. Macro F1	Base. Match F1	LLM Acc.	LLM Macro F1	LLM Match F1
Breakdown by Difficulty Pool						
Ordinary Pool (`hard=0`)	290	0.9310	0.9091	96.90%	**0.9647**	0.9524
Hard Pool (`hard=1`)	235	0.3717	0.0000	93.19%	**0.9296**	0.9167
Breakdown by Source Jurisdiction Pair						
DPD / EMA (Canada ↔ EU)	150	0.7366	0.6383	98.67%	**0.9864**	0.9844
DPD / FDA (Canada ↔ US)	116	0.6979	0.5625	97.41%	**0.9704**	0.9600
EMA / FDA (EU ↔ US)	259	0.7408	0.6176	92.28%	**0.9144**	0.8876

Qualitative Clinical Error Analysis. Inspecting all 25 test-set discrepancies (17 False Positives and 8 False Negatives) reveals four domain patterns:

1. **Cross-Border Licensing and Corporate Acquisitions (14 FP):** In 14 pairs with identical brand names and active substances (e.g. *Xolair* marketed by Novartis in the EU and Genentech in the US; *Doptelet* by SOBI vs. wholly owned subsidiary AkaRx; *Tekturna* by Novartis vs. acquirer LXO; *Nevanac* by Novartis vs. Harrow Eye; *Adasuve* by Ferrer vs. Nova Pneuma; *Zurampic* by Grünenthal vs. Ironwood), the LLM leveraged pretrained domain knowledge of licensing partnerships to predict `match` = 1, whereas annotators labeled them 0 due to differing surface manufacturer names (plus *Tevimbra* by BeOne Medicines Germany vs. Ireland, which is a gold-standard annotation error).

2. **Formulation Suffix Variants (3 FP, 1 FN):** Modified delivery formulations from the same innovator (*Natpar* vs. *Natpara*, *Alphagan* vs. *Alphagan P*, and *Afinitor Disperz* vs. *Afinitor* predicted 1 vs. GS 0; *Tobi Podhaler* dry powder inhaler vs. *TOBI* nebulizer solution predicted 0 vs. GS 1).

3. **Regional Co-Brands, Prodrugs, and Co-Formulations (3 FN):** *Copalia HCT* vs. *Exforge HCT* (distinct EU co-marketing brand names from Novartis), *Ivemend* (`fosaprepitant` intravenous prodrug) vs. *Emend* (`aprepitant` oral parent drug), and *Herceptin* (`trastuzumab`) vs. *Herceptin Hylecta* (`trastuzumab and hyaluronidase`).

4. **Gold-Standard Label Noise Corrected by the LLM (4 FN):** Four pairs labeled 1 in the manual gold standard are genuine non-matches where the LLM's 0 prediction is clinically correct: `AURO-LAMIVUDINE/ZIDOVUDINE` (dual therapy)

vs. EFAVIRENZ; LAMIVUDINE; ZIDOVUDINE (triple therapy), and three competing generic pairs (*Repaglinide Accord* vs. Aurobindo, *Pemetrexed Krka* vs. Shilpa, and *Tadalafil Lilly* vs. Rising).

4.5 Tri-Dataset Connected-Components Clustering

Pairwise entity resolution produces binary correspondence sets across DPD ↔ EMA, DPD ↔ FDA, and EMA ↔ FDA. To consolidate these pairwise links into unified multi-source medicine entities for Phase III fusion, we implemented graph-based transitive closure in src/identity_resolution/clustering.py using NetworkX connected components. Let $G = (V, E)$ be an undirected graph where vertices V are source-prefixed entity IDs (DPD_*, EMA_*, FDA_*) and edges E are positive pairwise match correspondences. Each connected component $C_k \subseteq V$ forms a unified medicine cluster assigned a deterministic identifier (CLUSTER_00001, ...).

Executing clustering.py on the exact normalized baseline correspondences across the three validated XML datasets produces **1,372 cross-dataset medicine clusters** spanning at least two jurisdictions, including **283 three-way clusters** present in Canada, the EU, and the US simultaneously (Table 14). This confirms that the integrated dataset exceeds the course threshold of 1,000 overlapping entities even before incorporating fuzzy and learned matcher correspondences. This count refines the preliminary exploratory profiling estimate of 1,407 reported in Section 2.3: whereas the exploratory estimate counted distinct synthetic keys across raw pre-consolidation CSV extracts, connected-components clustering operates directly on the deduplicated XML target datasets (where EMA was consolidated into 2,648 unique entities) and applies transitive graph closure to unify pairwise correspondences into coherent multi-source entity clusters.

Furthermore, when clustering is executed over the full candidate pool generated by the hybrid blocker ($\mathcal{C}_{\text{Qwen top-5}} \cup \mathcal{C}_{\text{Prefix-4}}$, 206,830 candidate pairs) using our hierarchical matcher with salt-form normalization and deterministic greedy 1:1 conflict resolution (--enforce-one-to-one), the yield of cross-dataset medicines increases to **2,343 clusters** (+70.8% gain over the exact baseline), with three-way international clusters increasing to **502** (+77.4% gain).

Table 14: Cross-jurisdictional medicine clusters formed via connected-components transitive closure: Baseline Exact vs. Candidate Correspondences (Deterministic Greedy 1:1)

Jurisdiction Combination	Exact Baseline	Share	Candidate Matches	Share
All 3 Sources (DPD + EMA + FDA)	283	20.63%	502	21.43%
Canada & US Only (DPD + FDA)	629	45.85%	1,388	59.24%
Canada & EU Only (DPD + EMA)	311	22.67%	314	13.40%
EU & US Only (EMA + FDA)	149	10.86%	139	5.93%
Total Overlapping Entities (≥ 2 Sources)	**1,372**	**100.00%**	**2,343**	**100.00%**

5 Phase III: Data Fusion

6 Conclusion

Sources

- U.S. Food and Drug Administration, *Drugs@FDA Data Files*. https://www.fda.gov/drugs/drug-approvals-and-databases/drugsfda-data-files
- European Medicines Agency, *Download medicine data*. https://www.ema.europa.eu/en/medicines/download-medicine-data
- Health Canada, *Drug Product Database*. https://www.canada.ca/en/health-canada/services/drugs-health-products/drug-products/drug-product-database.html